VLAN Configuration, Static Routing & Inter-VLAN Assignment

Comprehensive Technical Solutions & Packet Tracer Command Guide (Tasks 1 to 4)

Task 1: Creating 'Sales' (VLAN 10) & 'Support' (VLAN 20) in Cisco Packet Tracer

In Cisco Packet Tracer, adding a managed switch (such as a Cisco Catalyst 2960) allows you to partition a single physical switch into logical broadcast domains called Virtual LANs (VLANs). Below are the exact Cisco IOS CLI commands to configure VLAN 10 named **Sales** and VLAN 20 named **Support**:

Cisco IOS CLI Commands (Switch Configuration):

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Switch-Core

! --- Create VLAN 10 (Sales) ---
Switch-Core(config)# vlan 10
Switch-Core(config-vlan)# name Sales
Switch-Core(config-vlan)# exit

! --- Create VLAN 20 (Support) ---
Switch-Core(config)# vlan 20
Switch-Core(config-vlan)# name Support
Switch-Core(config-vlan)# exit
```

Explanation of Commands:

- `enable`: Enters Privileged EXEC mode.
- `configure terminal`: Enters Global Configuration mode.
- `vlan 10` / `vlan 20`: Creates the respective VLAN IDs in the switch's VLAN database (`vlan.dat`).
- `name Sales` / `name Support`: Assigns human-readable labels to identify each department's broadcast domain.

Task 2: Batch Switchport Assignment & Verification (`show vlan brief`)

To assign 3 ports (e.g., `Fa0/1` to `Fa0/3`) to the **Sales** VLAN and 2 ports (e.g., `Fa0/4` to `Fa0/5`) to the **Support** VLAN efficiently, we use the `interface range` command.

Switchport Assignment Commands:

```
! --- Assign Ports 1 to 3 to VLAN 10 (Sales) ---
Switch-Core(config)# interface range fastEthernet 0/1 - 3
Switch-Core(config-if-range)# switchport mode access
Switch-Core(config-if-range)# switchport access vlan 10
Switch-Core(config-if-range)# exit
```

```
! --- Assign Ports 4 to 5 to VLAN 20 (Support) ---
Switch-Core(config)# interface range fastEthernet 0/4 - 5
Switch-Core(config-if-range)# switchport mode access
Switch-Core(config-if-range)# switchport access vlan 20
Switch-Core(config-if-range)# end
Switch-Core# write memory
```

Verification Output (`show vlan brief` or `show vlan`):

```
Switch-Core# show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/6, Fa0/7, Fa0/8, Fa0/9...
10   Sales                             active    Fa0/1, Fa0/2, Fa0/3
20   Support                           active    Fa0/4, Fa0/5
1002 fddi-default                      act/unsup
1003 token-ring-default                act/unsup
1004 fddinet-default                   act/unsup
1005 trnet-default                     act/unsup
```

Port Summary Matrix:

VLAN ID	VLAN Name	Assigned Switch Ports	Operating Mode	Department / Usage
10	Sales	`FastEthernet 0/1 - 3` (3 Ports)	Access Mode	Sales Department PCs
20	Support	`FastEthernet 0/4 - 5` (2 Ports)	Access Mode	Technical Support PCs

Task 3: Connecting Two Routers with Static Routing Configuration

In this scenario, Router 1 (R1) connects to Router 2 (R2) via a point-to-point link (e.g., GigabitEthernet or Serial link on subnet `10.0.0.0/30`). R1 has LAN 1 (`192.168.10.0/24`) and R2 has LAN 2 (`192.168.20.0/24`).

Router 1 (R1) Configuration:

```
Router1> enable
Router1# configure terminal
Router1(config)# hostname R1

! --- Configure WAN Link to R2 ---
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

! --- Configure Static Route to reach LAN behind R2 (192.168.20.0/24 via R2 IP 10.0.0.2) ---
R1(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
R1(config)# end
R1# write memory
```

Router 2 (R2) Configuration:

```
Router2> enable
Router2# configure terminal
```

```
Router2(config)# hostname R2

! --- Configure WAN Link to R1 ---
R2(config)# interface gigabitEthernet 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

! --- Configure Static Route back to LAN behind R1 (192.168.10.0/24 via R1 IP 10.0.0.1) ---
R2(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
R2(config)# end
R2# write memory
```

Task 4: Inter-VLAN Simulation Analysis & Enabling Inter-VLAN Routing

Simulation Failure Analysis:
When a PC in the **Sales VLAN (VLAN 10)** attempts to ping a PC in the **Support VLAN (VLAN 20)** on the same Layer 2 switch, the ping **FAILS (Request Timed Out / Host Unreachable)**.

Why Does the Ping Fail?

- **Broadcast Domain Isolation:** VLANs break a physical switch into distinct, isolated Layer 2 broadcast domains. A switch will never forward frames between different VLAN IDs at Layer 2.
- **Subnet Separation:** Sales (VLAN 10) and Support (VLAN 20) reside in separate IP subnets. Moving traffic between different subnets requires a **Layer 3 routing device**.

Solution: Configure Inter-VLAN Routing ("Router-on-a-Stick")

To allow Sales and Support devices to communicate, connect a single trunk cable between the switch and a router interface (e.g., `Gi0/0/0` on Router), and configure sub-interfaces with IEEE 802.1Q encapsulation.

A. Switch Trunk Port Configuration:

```
Switch-Core(config)# interface gigabitEthernet 0/1
Switch-Core(config-if)# switchport mode trunk
Switch-Core(config-if)# exit
```

B. Router Sub-interface Configuration ("Router-on-a-Stick"):

```
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# no shutdown

! --- Sub-interface for Sales (VLAN 10) ---
Router(config)# interface gigabitEthernet 0/0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit

! --- Sub-interface for Support (VLAN 20) ---
Router(config)# interface gigabitEthernet 0/0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# end
```

✅ **Result:** Once "Router-on-a-Stick" (or a Layer 3 Switch with SVI interfaces enabled via `ip routing`) is configured, packets sent from Sales (VLAN 10) to Support (VLAN 20) are routed at Layer 3, and ping tests will succeed!